



Figure 3: Home page of the Metaverse Standards Forum

the world, namely, Meta, MSFT, Epic, and 33 others.

The forum will explore where the lack of interoperability is holding back metaverse deployment and how the work of Standards Developing Organisations (SDOs) defining and evolving needed standards may be coordinated and accelerated. The forum will not create standards *per se*. Instead, it will take on a more top-level, one might say ‘meta’, role by coordinating requirements and resources to foster the creation and evolution of standards for metaverse within standards organisations working in relevant domains.

Metaverse technologies and open source metaverse applications have predominantly been focused within the gaming space. However, as the concept, technologies and supporting infrastructure advance in concert with a general increase in public consciousness of the metaverse, we are seeing a rapid increase of metaverse based services and applications. This

includes applications for business, where there are numerous opportunities to rethink user interaction, distribution, and revenue channels, as well as how (and where) work is orchestrated and conducted.

There are several neighbouring yet distinct technologies that will need to be integrated and adopted into the metaverse for this to be fully realised. Amongst the most important of these are blockchain technologies, cloud infrastructure, Internet of Things (IoT), artificial intelligence, machine learning, robotics, and connectivity-based infrastructure such as satellite internet constellations and 5G/6G — all built on or predominantly open source.

Open source holds a critical role in the development of metaverse-aligned technologies and establishing the standards that will underpin the technology. There are numerous macro challenges that present significant obstacles to wider adoption of the metaverse -- for example, interoperability, identity, privacy,

security, sustainability, governance, and inclusivity. Open source software, projects and communities are essential for addressing and agreeing upon the best way to satisfy these challenges.

Blockchain technology will play an extremely vital role in surmounting these challenges and, consequently, many of the associated projects and communities of practice are open source. A prime example of this surrounds the projects focused on self-sovereign identity (SSI), which is an approach that gives individuals control of their digital identities and addresses the difficulty of establishing trust in an interaction. Given the transient and virtual/remote nature of human interaction in the metaverse, this question of identity, ownership and trust is at the heart of whether the metaverse is a safe, inclusive, and mutually beneficial place to work, socialise and play, or the very opposite.

The metaverse is increasingly becoming a viable place to work and to sell products and services, and